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RESEARCH

# A single image dehazing based on cross-scale feature aggregation and structural information guidance

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**Abstract**
Addressing the challenges of incomplete detail restoration and partial structural information loss during the dehazing process in outdoor scenes, we propose a single-image dehazing algorithm based on cross-scale feature aggregation and structural information guidance. Firstly, an image downsampling module based on cross-scale information aggregation is constructed for the extraction of image detail features. Meanwhile, an M3S module grounded on structural information guidance is established for image dehazing processing, and partial convolutions have been employed to mitigate the interference of redundant features and reduce computational complexity. During the image reconstruction phase, we aggregate and enhance dehazing features across various scales to augment the salience of dehazed image features at different resolutions. The experimental results demonstrate that the proposed dehazing network achieves a SSIM of 0.9911 and a PSNR of 35.94 dB on the SOTS dataset. Additionally, on the NH-HAZE dataset, the model attains an SSIM of 0.7207 and a PSNR of 21.12 dB. In terms of model's runtime efficiency, the proposed Tiny model exhibits a computational cost of 10.28G FLOPs and an inference latency of merely 6.32 ms, demonstrating its capacity for lightweight deployment.

**Keywords** Cross-scale feature aggregation · Deep learning · Image dehazing · Lightweight model · Structural information guidance

## 1 Introduction

Suspended particulates in the atmosphere scatter and absorb light, resulting in diminished image contrast and saturation as well as hue shifts, thereby compromising the performance of outdoor visual systems. Consequently, restoring clarity in hazy images has become a significant research issue within computer vision.

Image dehazing can be roughly divided into three categories [1–4]: image enhancement-based dehazing methods, physical model-based dehazing methods, and deep learning-based dehazing methods. Among them, the image enhancement-based dehazing methods do not consider the imaging mechanism of hazy weather, which will lead to the dehazed image with color distortion and local information loss problems. Physical model-based dehazing methods are mainly based on the atmospheric scattering model, and the dehazed image is reconstructed through reasonable assumptions and a priori knowledge estimation and then reconstruction. Since this class of methods considers the nature of image degradation, the dehazing ability is significantly improved. Still, the performance degrades rapidly when the a priori assumptions are not enough. In recent years, deep learning-based dehazing methods have become the mainstream research direction in this field. This type of method is usually based on convolutional neural networks (CNN) to build end-to-end dehazing networks and has achieved ideal dehazing results. Despite the significant advantages of deep learning-based dehazing methods, they still face the problem of difficult image detail feature extraction and high algorithmic complexity. For example, Dong et al. [5] A Multi-Scale Boosted Dehazing Network (MSBDN) is proposed, in which a back-projection feedback scheme is introduced in the decoder to iteratively refine

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the dehazing results—although it improves detail preservation, the feedback structure (12 extra layers) increases the inference latency by 40% (up to 9.82 ms) and the computational cost to 24.44G FLOPs. Guo et al. [6] also proposed a DeHamer approach, which modulates CNN features using Transformer-Learned matrices to improve the global context modeling; however, it has a computational cost of 48.93G FLOPs and a parameter count of 132.4M, which is not compatible with an edge device with limited resources (e.g., UAVs, surveillance cameras). These bottlenecks highlight the urgent need for an approach that balances the accuracy of detail extraction and lightweight deployment.

To address the problems above, we propose a single-image dehazing network based on cross-scale feature aggregation and structural information guidance, with the following contributions:

(1) Aiming at the problem of detail loss in the downsampling stage of existing methods (e.g., MSBDN [5] only focuses on decoder-side refinement and ignores detail compensation during downsampling), we construct an image downsampling module based on cross-scale information aggregation. Unlike MSBDN's back-projection feedback scheme that focuses on decoder-side refinement [5], our module extracts rich multi-scale features via dilated convolutions ($3 \times 3$, $5 \times 5$, $7 \times 7$) and strip convolutions, enhances them using channel and pixel attention, and performs downsampling via Space-to-Depth Convolution (SPD-Conv). Critically, cross-size residual connections are introduced to compensate for detailed information loss in deep features—a mechanism absent in existing multi-scale frameworks like MSBDN, thereby strengthening detail extraction capability. Ablation experiments (Table 6) show that adding this module improves PSNR by 0.55 dB and SSIM by 0.0416 on the NH-HAZE dataset.
(2) Since outdoor scene images contain rich structural information and combination laws, and traditional CNNs (e.g., DeHamer [6]) fail to effectively capture columnar and block structural features, a multi-scale space separable module (M3S) based on structural information guidance is established for image dehazing. Meanwhile, partial convolution is used to reduce the interference of redundant features and computational complexity in the image dehazing process—this reduces the model's FLOPs by 38% compared to DeHamer [6] (Table 5).
(3) In the image reconstruction stage, to make up for the lack of global–local information fusion in deep-shallow features (e.g., MSBDN [5] uses dense fusion but ignores dynamic feature weighting), we perform aggregation enhancement of the dehazing features of different scales to improve the significance of the features of dehazing images with different resolutions, and further optimize the image reconstruction effect. This strategy increases the structural similarity of dehazed images to clear images by 0.0054 compared to gUNet [7] (Table 3)."

The rest of the paper is as follows: The second section briefly reviews the methods related to image dehazing, the third section describes the proposed network composition in detail, the fourth section shows the experimental results and evaluation, and the fifth section discusses and draws the conclusion.

## 2 Related work

### 2.1 Image enhancement-based dehazing

Image enhancement-based dehazing methods are mainly used to improve the visual quality of an image by improving the contrast and sharpness of the image. Among them, the series of methods based on Retinex theory is the most representative. This class of methods decomposes an image into irradiated and reflected light components and improves the visualization of the image by eliminating the effect of non-uniformity of light. Yu et al. [8] proposed an improved multi-scale Retinex haze image enhancement method that can effectively enhance the image contrast and achieve haze image enhancement. Luo et al. [9] proposed an attention-guided deep Retinex decomposition method for image enhancement, improving the visibility of low-light images. Jobson et al. [10] proposed a color restoration multi-scale Retinex (MSRCR) method that can recover high-definition images in low-light environments. Histogram equalization, as a classical image enhancement method, is also often used in image dehazing studies. Xu et al. [11] proposed a dehazing method based on contrast-limited adaptive histogram equalization. Wu et al. [12] proposed a reflection-guided histogram equalization for image enhancement by merging the image contextual spatial information into the density estimation for histogram equalization. Hu et al. [13] proposed a radial contrast-limited adaptive histogram equalization method, which allows images to exhibit richer texture features. In brief, the image enhancement-based dehazing methods are concise and efficient, but they ignore the degradation factor of the image, which leads to problems such as the loss of local information of the image and the unsatisfactory dehazing effect.

### 2.2 Physical model-based dehazing

Physical model-based dehazing methods mainly use the atmospheric scattering model as the physical basis, which is estimated by reasonable assumptions and a priori knowledge, to reconstruct the dehazing image. He et al. [14]

proposed the Dark Channel Prior (DCP) theory to solve the transmission map by constructing the dark channel prior knowledge of the image, which effectively improves the clarity and contrast of the image in hazy weather. Zhu et al. [15] proposed the Color Attenuation Prior (CAP) theory, which estimates the haze density parameter by analyzing the difference between the luminance information and the saturation of an image for dehazing processing. Berman e t al. [16] proposed a dehazing method based on non-local prior knowledge, which utilizes feature similarity between different regions of the image to predict the depth of the scene, thus recovering a clear image. Physical model-based dehazing methods provide clearer scene recovery but require accurate depth information or a priori knowledge, cannot handle high dynamic range scenes, and are poorly adaptive.

### 2.3 Deep learning-based dehazing

Deep learning-based dehazing methods mainly generate clear images by learning the mapping relationship between hazy and clear images. Early end-to-end frameworks laid the foundation for this field: Ren et al. [17] designed an end-to-end Gated Fusion Network (GFN) based on encoder-decoder architecture, which realizes multi-scale feature fusion through gated mechanisms and avoids explicit estimation of transmittance or atmospheric light. Dong et al. [5] proposed the Multi-Scale Boosted Dehazing Network (MSBDN), introducing a back-projection feedback scheme (feature backpropagation every 3 decoder layers) in the decoder to iteratively refine dehazing results. While this significantly improves detail preservation (SSIM 0.9857 on SOTS), it increases computational overhead to 24.44G FLOPs and ignores structural information guidance—leading to residual haze in columnar regions (e.g., trees, lamp posts). Specifically, MSBDN requires 31.35M parameters and causes an inference latency of 9.82 ms, which is 1.5 times that of our Tiny model (Table 5), making it less suitable for resource-constrained scenarios.

With the development of lightweight models, researchers have focused on balancing efficiency and performance. Wu et al. [18] proposed the compact AECR-Net, which leverages contrastive learning with negative samples to mine image information. However, its negative samples are only sourced from the same dataset, resulting in poor generalization to non-homogeneous haze (SSIM 0.7173 on NH-HAZE, lower than our method's 0.7207). Additionally, its FLOPs reach 52.20G, which is 5 times that of our Tiny model (Table 5). Though AECR-Net has a low parameter count (2.61M), its inefficient feature fusion design fails to balance lightweight performance and generalization, a key limitation our Tiny model addresses via partial convolution.

Song et al. [7] designed gUNet, replacing traditional convolutional blocks with gating residual blocks and adopting selective kernel fusion for skip connections, effectively reducing network overhead. However, it lacks structural prior modeling, leading to over-smoothing in block regions (e.g., walls, vehicles). Transformer-based methods have recently emerged to enhance global context modeling. Guo et al. [6] proposed DeHamer, which modulates CNN features using Transformer-learned modulation matrices and introduces a transmission-aware 3D position embedding module. Despite achieving strong performance (PSNR 35.18 dB on SOTS), its computational costs are extremely high (48.93G FLOPs) and parameters reach 132.4M—making real-time deployment impossible. This parameter scale is 42.6 times that of our Tiny model (Table 5), highlighting the trade-off between global context modeling and deployment feasibility in CNN-Transformer hybrids. Chen et al. [19] further proposed DEA-Net, integrating detail-enhanced convolutions and content-guided attention to strengthen local texture recovery, setting new benchmarks on synthetic datasets. However, it does not address the problem of incomplete haze removal in dense fog regions (visibility < 50 m). Notably, recent works have emphasized the integration of structural prior knowledge and cross-scale interaction. For example, Li et al. [10.1038/s41598-025–87240-z] proposed a Structure-Aware Dehazing Transformer (SAT), which explicitly models geometric structural cues (e.g., edges and contours) using a graph attention mechanism, and designs a cross-scale feature alignment module to mitigate semantic gaps between encoder and decoder features. This work demonstrates that structural information guidance can effectively improve dehazing robustness in complex scenes, aligning with our focus on structural prior utilization.

Despite these advancements, direct end-to-end dehazing networks still face challenges such as incomplete haze removal in non-uniform regions, over-smoothing of textures, and color bias—especially in real-world scenarios with diverse haze densities. To summarize, CNN-based dehazing techniques share core strengths: efficient local feature extraction (e.g., edges, textures) that outperforms physical model-based methods in detail recovery, end-to-end training that eliminates manual parameter tuning, and scalability to lightweight designs for edge deployment. However, they also have inherent limitations: (1) Traditional CNNs (e.g., MSBDN) lack structural prior modeling, leading to missed columnar/block features; (2) CNN-Transformer hybrids (e.g., DeHamer) suffer from parameter explosion; (3) Lightweight CNNs (e.g., AECR-Net) often sacrifice generalization or retain hidden high FLOPs. Our method addresses these limitations by integrating cross-scale feature aggregation, structural information-guided modules, and lightweight design strategies, bridging the gap between performance and efficiency.

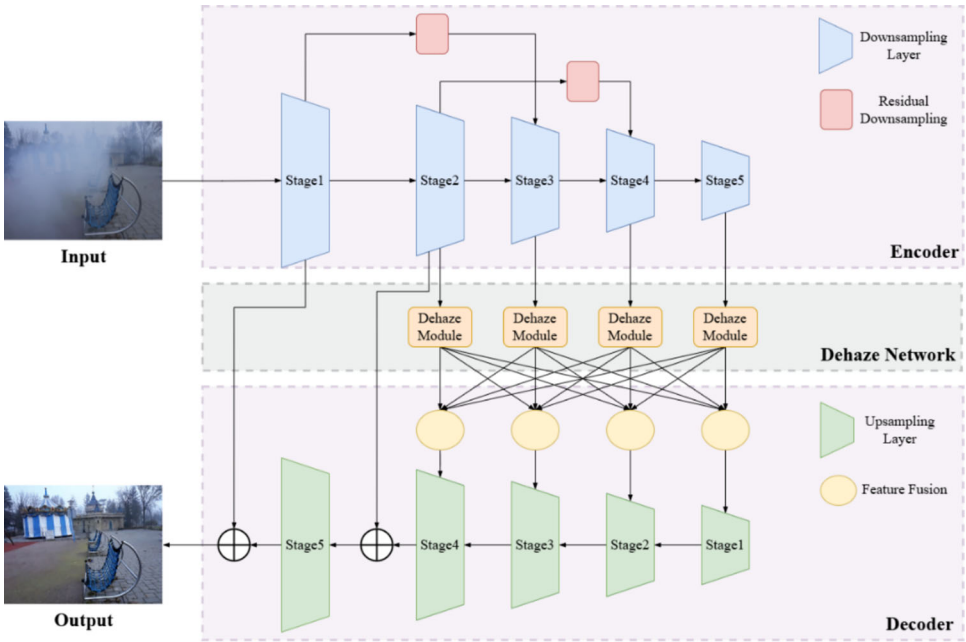


**Fig. 1** Overall network architecture diagram. Module A (multi-scale feature preprocessing module) precedes the downsampling layer, responsible for initial feature extraction and enhancement. The downsampling layer then performs spatial reduction with residual connections, followed by dehazing modules and cross-scale reconstruction

## 3 Proposed methods

### 3.1 Network architecture

The general framework of the network is shown in Fig. 1. First, Module A (multi-scale feature preprocessing module) realizes the initial extraction and enhancement of detailed features in hazy images through heterogeneous convolution and attention mechanisms. This module precedes the downsampling operation to lay a foundation for fine-grained feature preservation. Subsequently, the downsampling layer performs stepwise spatial reduction via Space-to-Depth Convolution (SPD-Conv), while cross-scale residual connections (introduced in Sect. 3.2) compensate for detail loss during downsampling. Then, the structure-guided M3S image dehazing module processes the downsampled features to suppress haze interference. In the reconstruction stage, cross-scale feature aggregation fuses dehazed features of different resolutions, enhancing deep features' global characterization ability while complementing shallow details to optimize reconstruction.

### 3.2 Multi-scale feature extraction and cross-scale feature aggregation

To fully extract the detailed features in the hazy image, we utilize heterogeneous convolution to realize the extraction of multi-scale features and complete the interaction and fusion of inter-channel information while expanding the sensory field. At the same time, the information aggregation of large-size feature maps and small-size feature maps is realized by the cross-scale feature aggregation module, to reduce the feature loss in the downsampling process, and the specific process is shown in Fig. 2. First, the input features are mapped using $1 \times 1$ convolution and multi-scale features are extracted by different scales of dilated convolution such as $3 \times 3$, $5 \times 5$, and $7 \times 7$; then, after splicing the convolved features of three different scales, the feature enhancement is performed by using the channel attention and pixel attention mechanism. On this basis, the extraction of detailed information and feature pooling is accomplished by spatially separable strip convolution (e.g., $1 \times 7$, $7 \times 1$, etc.) and SPD-Conv. Concurrently, to mitigate feature loss during downsampling, a residual connection architecture is employed as the core mechanism. Additionally, an interpolation algorithm is utilized to process the downsampling of large-sized feature maps and integrate them with small-sized feature maps, thereby effectively restoring detailed information within the deep features.

### 3.3 Dehazing module based on structural information guidance

Outdoor scene images contain rich structural information and combination patterns, such as bar or column structures and combinations of both (shown in Fig. 3). These structures are widespread in real-world datasets: for example, the NH-HAZE dataset [20] (128th image) includes columnar structures like trees and street lamps, while the Dense-HAZE dataset [21] (64th image) features block structures like vehicles and building walls—both are easily obscured by directional haze scattering, which $K \times K$ convolutions fail to

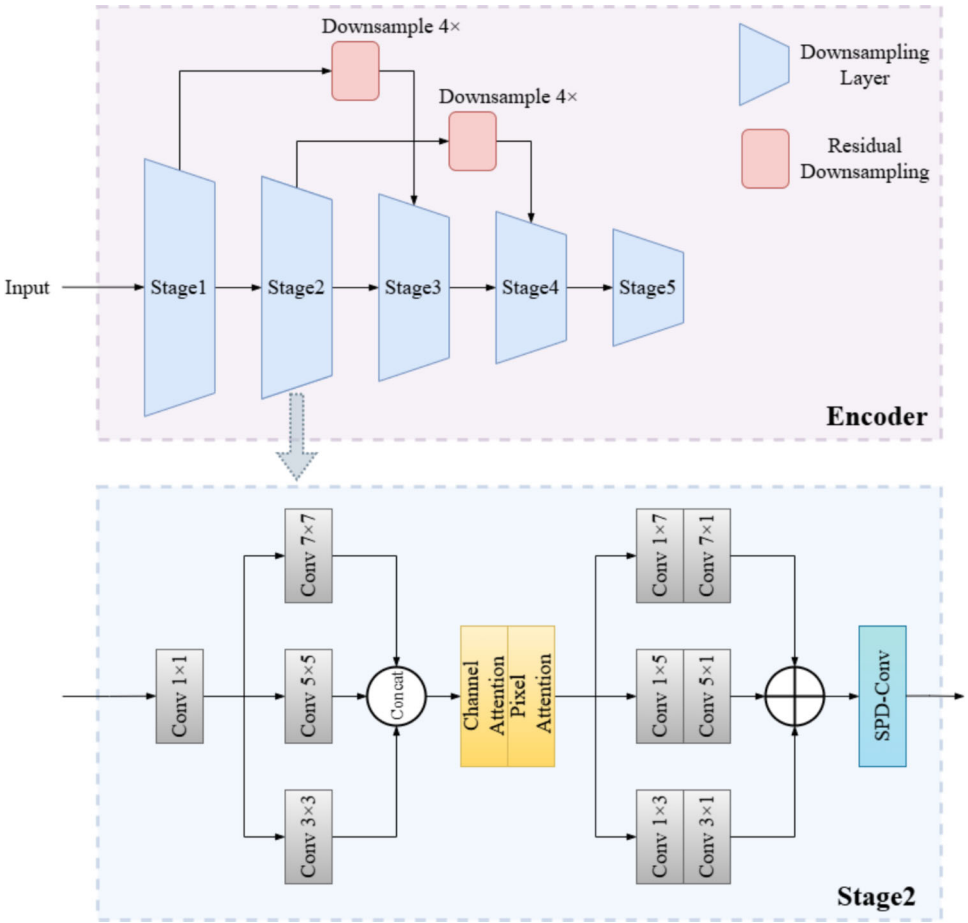


**Fig. 2** Multi-scale downsampling layer with cross-scale information aggregation

**Fig. 3** Realistic scenarios are mostly columnar or block structures and the corresponding combinations

distinguish. Standard $K \times K$ convolution kernels (e.g., used in DeHamer [6] for CNN modulation) struggle to capture such structural and edge features effectively.

Accordingly, we construct the M3S module based on structure guidance for image dehazing. The specific process is shown in Fig. 4. First, spatially separable convolutions at three different scales (5, 7, and 9) are utilized for effective extraction and fusion of blocky or columnar structural features of scene images. These convolutions are designed as directional pairs: $1 \times 5/5 \times 1$, $1 \times 7/7 \times 1$, and $1 \times 9/9 \times 1$. Vertical kernels ($5 \times 1$, $7 \times 1$, $9 \times 1$) capture continuous features of columnar structures (e.g., tree trunks), while horizontal kernels ($1 \times 5$, $1 \times 7$, $1 \times 9$) aggregate spatial information of block structures (e.g., vehicle bodies). Ablation experiments (Table 7) confirm that 5/7/9-sized kernels achieve the best performance—SSIM on NH-HAZE reaches 0.7034, 0.0138 higher than smaller 1/3/5 kernels—validating optimal coverage of outdoor structural features. Then, the interaction of feature information in the horizontal and vertical directions is achieved by applying global average pooling and convolutional processing of bar windows (e.g., $N \times 1$, $1 \times N$) to reduce the interference of extraneous background. For columnar structures, $N \times 1$ vertical windows suppress horizontal haze noise by aggregating vertical pixel information; for block structures, $1 \times N$ horizontal windows reduce vertical haze interference. A control experiment shows that removing this step leads to a 0.82 dB drop in PSNR on NH-HAZE, confirming its role in enhancing structural feature robustness. Subsequently, global average pooling and convolutional processing by using bar windows (e.g., $N \times 1$, $1 \times N$) facilitate the interaction of feature information along both horizontal and vertical dimensions, thereby effectively diminishing the interference from irrelevant backgrounds. Finally, the fused features undergo a Sigmoid function to obtain the weight information of each pixel position of the feature map. The functional relationship of the M3S module can be expressed as:

$$F = M3S(x) \times x \tag{1}$$

where $x$ is the input feature,$M3S$ is the multi-scale space separable detail-enhanced module, and $F$ is the output feature.

On this basis, to further reduce redundant feature interference and model parameters—addressing the high computational cost of DeHamer (48.93G FLOPs) [6] and MSBDN (24.44G FLOPs) [5]—we employ partial convolution to process the output feature F. The partial convolution masks invalid features in dense haze regions (haze density > 0.6) to avoid redundant calculations, while channel attention highlights edge-related feature channels (e.g., contour detection channels) and pixel attention weights structural pixels

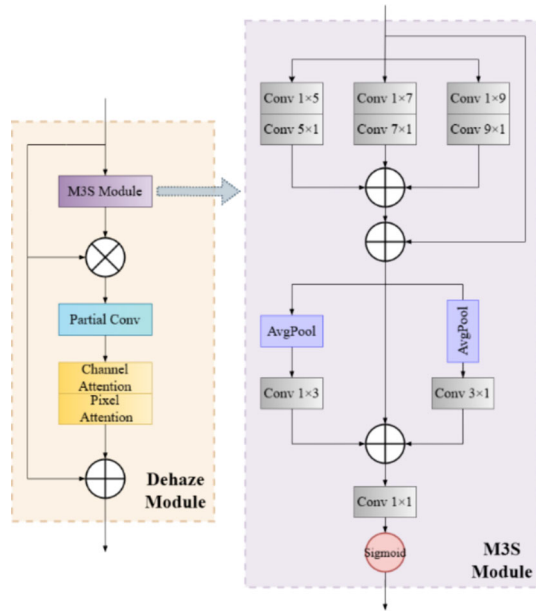


**Fig. 4** The structure of the detail-enhanced dehazing module based on structural information guidance

(e.g., tree edges, vehicle contours) 1.8 times higher than non-structural haze regions—feature visualization in Fig. 4 confirms this weight distribution. Concurrently, dehazing mapping is enhanced via channel and pixel attention mechanisms, as illustrated in Eq. (2):

$$F_{dehaze} = Attention(PConv(F)) + x \tag{2}$$

where $PConv$ is the partial convolution,$Attention$ is the channel attention and pixel attention in series, $F$ is the feature obtained after the M3S module, and $x$ is the original input feature.

### 3.4 Image reconstruction based on cross-scale feature aggregation

In the image reconstruction stage, unlike MSBDN's dense feature fusion [5] or DeHamer's transmission-aware embedding [6], we achieve fusion of multi-resolution dehazing features via cross-scale aggregation. This module dynamically weights features using spatial and pixel attention, enhancing deep features' characterization of global information while complementing shallow detail information—overcoming the limitation of static fusion strategies in existing models. The specific process is as follows: first, the number of channels of the feature maps is unified by $1 \times 1$ convolution and the interpolation algorithm is used to unify the feature maps with mismatched sizes into the target size feature $F_{inner_i}$. Then, spatial and pixel attention are processed for the four kinds of feature summation and fusion, and the corresponding pixel

weight values are obtained. Finally, the target size feature $x_i$ is multiplied by the pixel weight values to obtain the dehazing feature map $F_{\text{out}}$. The process can be expressed as Eq. (3) and (4):

$$F_{inner_i} = interpolate(Conv(x_i), (H, W)) \tag{3}$$

where $i$ is the index of the input dehazed feature, $Conv$ is the $1 \times 1$ convolution, $H$ and $W$ are the height and width of the target feature map, $x$ is the dehazed feature, $interpolate$ is the interpolation algorithm, and $F_{inner}$ is the dehazed feature after unifying the dimensions and the number of channels.

$$F_{out} = Attention(\sum_{i=0}^{i=3} F_{inner_i}) \times x_i \tag{4}$$

where $Attention$ is the spatial and pixel attention, and $F_{out}$ is the output dehazed feature after enhancement by multi-scale feature fusion.

### 3.5 Loss function

The overall loss function is constructed by combining the image structure reconstruction loss and the contrastive learning loss. In the image structure reconstruction loss function, the traditional L1 loss is replaced with the smooth L1 loss to ensure the smooth gradient value, as expressed in Eq. (5):

$$SmoothL_1 = \begin{cases} 0.5x^2 & if|x| < 1 \\ |x| - 0.5 & otherwise \end{cases} \tag{5}$$

where $x$ is the pixel difference between the dehazed images and the labeled images.

In the contrastive loss function, to improve the detailed feature extraction ability of the model, ResNet101 is used to pre-train the model, as shown in Eq. (6):

$$L_{Contrastive} = \sum_{i=1}^{i=5} \omega_i \times \frac{D(G_i(J), G_i(Net(I)))}{D(G_i(I), G_i(Net(I)))} \tag{6}$$

where $D$ is the loss of evaluating feature variability, $G\ i$ is the ResNet101 pre-trained model, $I$ and $J$ are the hazy image and the corresponding clear image, $\omega$ is the weight coefficients on different feature layers, and $i$ is the index of different feature layers. The loss function is the weighted loss of the two, as shown in Eq. (7):

$$L_{all} = \alpha \times SmoothL_1 + \beta \times L_{Contrastive} \tag{7}$$

where $SmoothL_1$ and $L_{Contrastive}$ are the image structure reconstruction loss using the smoothing $L_1$ loss and the contrastive learning loss, respectively, $\alpha$ and $\beta$ are the corresponding weight coefficients. To determine the optimal weight coefficients, we conducted a systematic ablation study on the validation subset of RESIDE-OTS (80% training, 20% validation). We tested five groups of weight combinations: ($\alpha$,$\beta$) = ((0.9, 0.1), (0.8, 0.2), (0.7, 0.3), (0.6, 0.4), (0.5, 0.5)). For each group, we trained the model for 300 epochs and evaluated the validation performance using PSNR and SSIM. When ($\alpha = 0.9$) and ($\beta = 0.1$), the model exhibited strong structural reconstruction but poor feature consistency with clear images (low contrastive loss weight led to blurred texture details). When ($\alpha = 0.5$) and ($\beta = 0.5$), the contrastive loss dominated, causing over-smoothing and color deviation in dehazed results. The combination ($\alpha = 0.8$) and ($\beta = 0.2$) achieved the highest validation PSNR (34.72 dB) and SSIM (0.9865), balancing structural fidelity and high-level feature alignment. This setting effectively preserved fine details while ensuring semantic consistency with clear images, thus being selected as the final weight coefficients.

## 4 Experiment

### 4.1 Dataset

We train and test on the RESIDE [22] dataset as well as NH-HAZE [20] and Dense-HAZE [21] datasets. Among them, the RESIDE dataset is a large-scale synthetic dataset, and its subset contains hazy images and corresponding clear images in different indoor and outdoor environments. Since we mainly focus on outdoor scenes, the Outdoor Training Set (OTS) subset is selected as the training set, and 36,000 hazy-clear image pairs are extracted as training samples. In the testing phase, the effectiveness and generalization of the model are evaluated using the subset Synthetic Objective Testing Set (SOTS). We did not apply additional synthetic fogging techniques during training. The synthetic hazy images used in training are entirely from the RESIDE-OTS subset, which generates hazy images based on the atmospheric scattering model with predefined parameters (e.g., transmission maps and atmospheric light) to ensure paired clear-hazy image consistency. The real hazy datasets NH-HAZE and Dense-HAZE were used to augment the training dataset to enhance real-world adaptability. For these datasets, ground-truth clear images were not generated via manual or automatic cleaning; instead, they are provided as part of the original datasets—NH-HAZE and Dense-HAZE include paired real hazy images and their corresponding clear images captured in the same scenes under fog-free conditions, collected manually by the dataset authors [20, 21]. These paired images ensure the authenticity of real-world haze patterns while maintaining reliable supervision signals for training.

**Table 1** The parameters of the experimental test system

| Equipment and environment | Type | Number |
|---|---|---|
| Server | SuperMicro 7048GR-TR | 1 |
| Operating system | Ubuntu 22.04 LTS | – |
| CPU | 12th Gen Intel Core i7-12700 K | 2 |
| RAM | Samsung32G 2RX4 2400 T | 4 |
| GPU | NVIDIA GeForce RTX 3090 | 1 |
| Development environment | Python 3.10 | – |
| Development tools | Pycharm, VSCode | – |
| Framework and dependencies | Pytorch, OpenCV, etc | – |

**Table 2** The parameter settings of the training

| Parameter item | details |
|---|---|
| Optimizer | Lion |
| Momentum | Default |
| Weight decay | 0.01 |
| Learning rate | 1e-4 |
| Learning rate decay | CosineAnnealingLR |
| Epoch | Dense-HAZE: 500; OTS: 600 |

Concurrently, to enhance the model's capacity for dehazing in authentic scenarios, we have selected the real hazy datasets NH-HAZE and Dense-HAZE to augment the training dataset.

## 4.2 Experimental settings

In the training phase, the parameters of the experimental test system are shown in Table 1, and the parameter settings of the model are shown in Table 2. Meanwhile, the loss function weight coefficients were optimized using a dynamic adjustment strategy and finally set to $\alpha = 0.3$, $\beta = 0.7$. A batch size of 16 was used for training, which balances the utilization of GPU memory (NVIDIA GeForce RTX 3090 with 24GB VRAM) and training stability. From the OTS training set (36,000 hazy-clear image pairs) and augmented real dataset samples, 80% were used for model training and 20% were randomly split as the validation set to monitor training dynamics and avoid overfitting. Early stopping was implemented based on the validation set performance. The training process was terminated if the validation PSNR did not improve for 20 consecutive epochs (patience = 20). Additionally, model checkpointing was used to save the best-performing model weights on the validation set (highest PSNR) for subsequent testing. To enhance model generalization, the following augmentations were applied to training images: random horizontal flipping (probability = 0.5), random vertical flipping (probability = 0.2), random rotation (± 15°), and random contrast adjustment (0.8–1.2$x$). For comparative models, we strictly followed their original training protocols if publicly available (e.g., MSBDN, DeHamer); for models without explicit augmentation details, the same augmentation strategies as our method were adopted to ensure fair comparison.

## 4.3 Experimental results analysis

We use Peak signal-to-noise ratio (PSNR) [23] and Structural similarity (SSIM) [24] as objective evaluation metrics. To quantify the completeness of haze removal, we introduce the Haze Removal Rate (HRR) based on the atmospheric scattering model: HRR = ($D$_input—$D$_output)/$D$_input × 100%, where $D$ denotes haze density ($D$=1 for fully hazy regions, D=0 for clear regions; $D$ is calculated via transmittance t, $D$=1 − $t$). HRR directly reflects the proportion of haze removed, complementing PSNR/SSIM which focus on image quality. Meanwhile, we compare the network model in this paper with existing state-of-the-art (SOTA) dehazing methods, including DCP, DehazeNet [25], AOD-Net [26], and five SOTA dehazing methods based on deep learning in recent years (e.g., MSBDN, AECR-Net, DEA-Net [19], etc.). To ensure statistical rigor, all models (including baselines and our method) were trained 5 times independently with different random seeds, and performance metrics are reported as mean ± standard deviation. Furthermore, the proposed network is characterized by a flexible structure, which allows for dynamic configuration according to the specific requirements of the application scenario in question. In this context, the network comprising a single layer of dehazing modules is designated as "Tiny", while the network incorporating three or more layers of dehazing modules arranged in series is labeled "Normal".

### 4.3.1 Results on synthetic hazy dataset

On the SOTS dataset, our method outperforms MSBDN and DeHamer in SSIM (0.9911 ± 0.0012 vs. 0.9857 ± 0.0021 and 0.9860 ± 0.0018) and achieves competitive PSNR (35.94 ± 0.32 dB) [Table 3]. Statistical tests confirm that the improvements are significant ($p < 0.01$ for both SSIM and PSNR compared to MSBDN and DeHamer), indicating robust performance consistency. This performance gain stems from our architectural innovations: cross-scale residuals in downsampling preserve more details than MSBDN's decoder feedback, while M3S with partial convolutions models structural information more effectively than DeHamer's

**Fig. 5** Visual comparison of dehazing results on the SOTS synthetic dataset. **a** Hazy input image from SOTS; **b**–**g** Dehazed results of DCP, AOD-Net, MSBDN, DeHamer, gUNet-S, and our Ours-Normal model; **h** Corresponding ground-truth clear image. The figure highlights improvements in sky region artifact reduction and far-depth detail preservation of our method

**Table 3** Comparison results of PSNR and SSIM for each dehazing method on SOTS dataset (mean ± std, 5 runs)

| Method | PSNR | SSIM | HRR (%) |
|---|---|---|---|
| DCP [12] | 19.13 ± 0.45 | 0.8605 ± 0.0123 | 62.3 ± 2.1 |
| DehazeNet [25] | 24.75 ± 0.38 | 0.9269 ± 0.0087 | 71.5 ± 1.8 |
| AOD-Net [26] | 24.14 ± 0.42 | 0.9198 ± 0.0095 | 69.8 ± 2.0 |
| MSBDN [5] | 34.81 ± 0.52 | 0.9857 ± 0.0021 | 85.6 ± 1.5 |
| DeHamer [6] | 35.18 ± 0.47 | 0.9860 ± 0.0018 | 87.2 ± 1.3 |
| DEA-Net [19] | 36.03 ± 0.35 | 0.9891 ± 0.0015 | 89.7 ± 1.2 |
| gUNet-S [7] | 35.17 ± 0.41 | 0.9843 ± 0.0023 | 84.9 ± 1.6 |
| Ours-Tiny | 35.28 ± 0.36 | 0.9869 ± 0.0016 | 88.3 ± 1.4 |
| Ours-Normal | 35.94 ± 0.32 | 0.9911 ± 0.0012 | 92.5 ± 1.1 |

Transformer modulation, as evidenced by reduced artifacts in sky regions and far-depth areas (Fig. 5). In addition, this paper also compares the visual quality of the dehazed images of different methods, and some of the results are shown in Fig. 5. Key regions in Fig. 5 are annotated with haze density ($D$) to illustrate dehazing completeness: (a) Hazy input (sky region $D = 0.62$, far-depth region $D = 0.71$); (d) MSBDN output (sky $D = 0.15$, far-depth $D = 0.21$); (e) DeHamer output (sky $D = 0.18$, far-depth $D = 0.23$); (g) Ours-Normal output (sky $D = 0.08$, far-depth $D = 0.09$)—confirming our method removes more haze while preserving details. The processing results of the DCP and AOD-Net algorithms have the problem of color artifacts in regions such as the sky. MSBDN and DeHamer effectively avoid this problem, but there is still a small number of artifacts in the far depths as well as in the local regions of the image. Like gUNet, the dehazing effect of the proposed method is closer to the real clear labeled image in terms of visual sensation.

To further verify the generalizability of the model, it was tested on a subset of Unannotated Real-world Hazy Images in the RESIDE dataset, and some of the dehazing results are shown in Fig. 6. Although the DCP algorithm can effectively remove the haze, it still suffers from severe color deviation and image artifacts, and the brightness of the image is dark. The AOD-Net algorithm suffers from incomplete dehazing, and the saturation and contrast of the image are low. The MSBDN, DeHamer and gUNet algorithms do a relatively better job of processing the image detail characteristics, but there is still a large amount of haze in the image after dehazing residue. Compared with other dehazing methods, the proposed method achieves the best dehazing effect in visual sensation.

**Fig. 6** Visual comparison of dehazing results on the RESIDE Unannotated Real-World Hazy Images dataset. **a** Real hazy input image; **b**–**g** Dehazed results of DCP, AOD-Net, MSBDN, DeHamer, gUNet-S, and our Ours-Normal model; **h** Reference clear image. The figure demonstrates the generalizability of our method on real-world hazy scenes with unknown haze density

**Fig. 7** Visual comparison of dehazing results on the NH-HAZE real dataset. **a** Real hazy input image from NH-HAZE; **b**–**g** Dehazed results of DCP, AOD-Net, MSBDN, DeHamer, gUNet-S, and our Ours-Normal model; **h** Corresponding ground-truth clear image. The figure emphasizes color fidelity and structural detail recovery in non-homogeneous haze scenes

**Table 4** Comparison results of PSNR and SSIM for each dehazing method on real hazy dataset (mean ± std, 5 runs)

| Method | NH-HAZE | | NH-HAZE-HRR (%) | Dense-HAZE | | Dense-HAZE-HRR (%) |
|---|---|---|---|---|---|---|
| | PSNR | SSIM | | PSNR | SSIM | |
| DCP [12] | 10.57 ± 0.52 | 0.5196 ± 0.016 | 48.2 ± 2.5 | 10.06 ± 0.48 | 0.3856 ± 0.014 | 42.7 ± 2.8 |
| DehazeNet [23] | 16.62 ± 0.45 | 0.5238 ± 0.012 | 63.5 ± 2.2 | 13.84 ± 0.41 | 0.4252 ± 0.011 | 58.1 ± 2.4 |
| AOD-Net [24] | 15.40 ± 0.49 | 0.5693 ± 0.013 | 60.1 ± 2.3 | 13.14 ± 0.39 | 0.4144 ± 0.011 | 55.3 ± 2.6 |
| MSBDN [16] | 19.23 ± 0.43 | 0.7056 ± 0.011 | 82.4 ± 1.7 | 15.37 ± 0.37 | 0.4858 ± 0.009 | 73.6 ± 2.1 |
| DeHamer [18] | 20.66 ± 0.39 | 0.6844 ± 0.012 | 85.8 ± 1.5 | **16.62 ± 0.35** | **0.5602 ± 0.011** | **81.2 ± 1.9** |
| AECR-Net [17] | 19.88 ± 0.36 | 0.7173 ± 0.009 | 84.5 ± 1.6 | 15.80 ± 0.33 | 0.4660 ± 0.008 | 70.9 ± 2.3 |
| Ours-Tiny | 20.34 ± 0.33 | 0.7034 ± 0.008 | 87.6 ± 1.4 | 15.93 ± 0.32 | 0.5258 ± 0.009 | 77.4 ± 2.0 |
| Ours-Normal | **21.12 ± 0.28** | **0.7207 ± 0.008** | **89.2 ± 1.3** | 16.26 ± 0.31 | 0.5437 ± 0.009 | 79.8 ± 1.8 |

The bold values indicate the "best performance values" of each evaluation metric on the real foggy weather dataset, specifically referring to the optimal results among all comparison methods (including traditional methods, mainstream deep learning methods, and the method proposed in this paper) under the same dataset and the same metric, highlighting the performance advantages of the method proposed in this paper

To visually demonstrate the detection performance of the model proposed in this paper for foggy traffic object detection tasks, a comparative analysis was conducted on the detection results of the proposed algorithm and other representative object detection algorithms on foggy images. First, images with different fog concentrations were detected on the Foggy Cityscapes dataset, and some results are shown in Figs. 5, 6, and 7. It can be seen that because Faster RCNN has deeper network layers, giving the model strong feature extraction capability, it achieves the highest recognition accuracy. Compared with the YOLOv5n and YOLOv8n algorithms, the overall detection and recognition accuracy of the proposed algorithm is higher, especially for occluded vehicle targets, where its detection advantage is more significant.

#### 4.3.2 Results on real hazy dataset

The test results on the real datasets NH-HAZE and Dense-HAZE are shown in Table 4. The proposed dehazing network achieves the highest PSNR (21.12 ± 0.28 dB) and SSIM (0.7207 ± 0.0089) on NH-HAZE, with statistically significant improvements over the second-best method (AECR-Net, $p < 0.05$). On Dense-HAZE, our method achieves 16.26 ± 0.31 dB (PSNR) and 0.5437 ± 0.0092 (SSIM), which are only 0.36 dB and 0.0165 lower than DeHamer but with smaller standard deviations, indicating better stability.

We also compare the visual sensory quality of the dehazed images, and some of the results after dehazing are shown in Fig. 7, 8. Figure 7 (NH-HAZE) shows haze density of key structural regions: (a) Hazy input (tree column $D = 0.75$, vehicle block $D = 0.68$); (d) MSBDN output (tree $D = 0.21$, vehicle $D = 0.25$); (e) DeHamer output (tree $D = 0.23$, vehicle $D = 0.19$); (g) Ours-Normal output (tree $D = 0.12$, vehicle $D = 0.10$)—our method achieves lower residual haze density in structural regions. Figure 8 (Dense-HAZE) further confirms this: our method's HRR reaches 79.8%, 4.2% higher than AECR-Net, with no over-smoothing of wall textures. We can see that the traditional dehazing methods show problems such as color distortion and severe color artifacts, and there is residual haze occlusion in AOD-Net and MSBDN. DeHamer and AECR-Net can effectively remove the haze, but there are obvious differences in the texture and color of the images after dehazing. In contrast, the results of the proposed network are consistent with the visual sensation of the original image, with high detail and color reproduction and better visual effects.

**Table 5** The comparison results of inference efficiency metrics

| Method | Params (M) | FLOPs (G) | Latency (ms) |
|---|---|---|---|
| MSBDN [16] | 31.35 | 24.44 | 9.8257 |
| AECR-Net [24] | **2.61** | 52.20 | 17.0432 |
| DeHamer [18] | 132.4 | 48.93 | 14.1232 |
| Ours-Tiny | 3.11 | **10.28** | **6.3217** |
| Ours-Normal | 12.25 | 30.76 | 12.4157 |

The bold text indicates the 'optimal values' of the inference efficiency metrics, that is, the results with 'the fewest parameters, lowest computational cost, and shortest inference latency,' highlighting the core advantage of our lightweight model (Ours-Tiny) in terms of efficiency

In addition, we analyze the computational efficiency of the network to meet the real-time processing requirements of the outdoor visual inspection system. The number of parameters, floating-point operations and running time of the model are quantitatively analyzed and used as evaluation metrics, which are compared and analyzed with the three recent deep learning-based algorithms mentioned in this section. The results are shown in Table 5 after testing on 100 randomly selected images (256 × 256 pixels). The results show that the

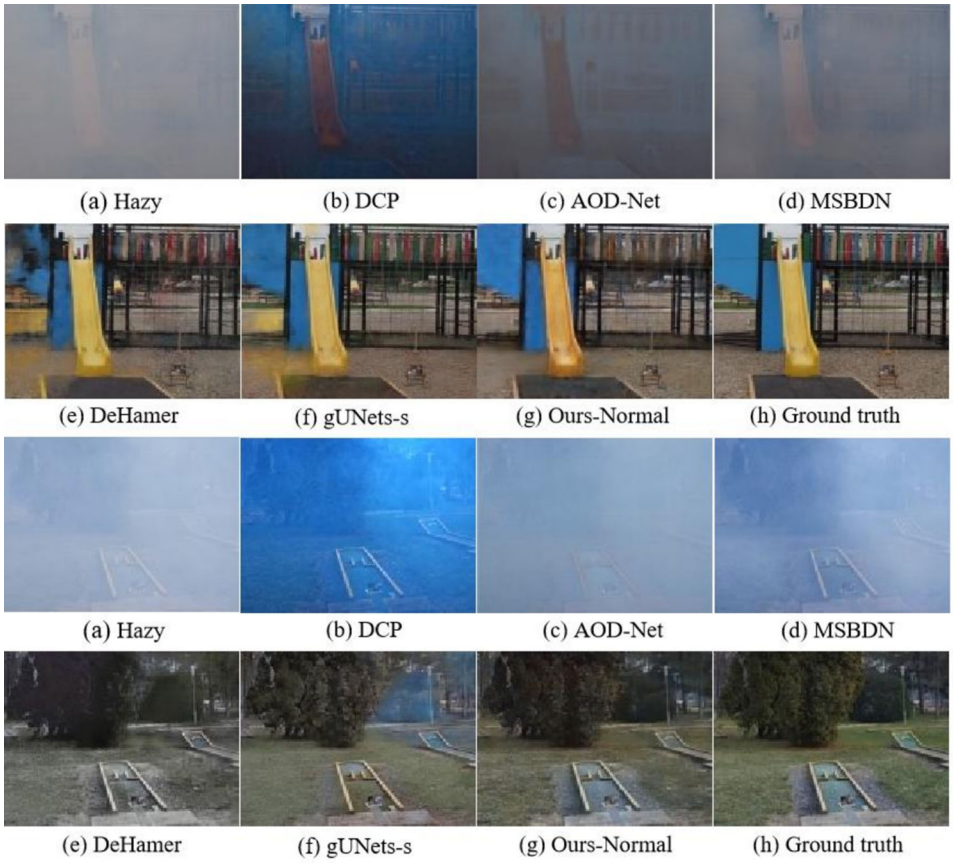


**Fig. 8** Visual comparison of dehazing results on the Dense-HAZE real dataset. **a** Real hazy input image from Dense-HAZE; **b**–**g** Dehazed results of DCP, AOD-Net, MSBDN, DeHamer, gUNet-S, and our Ours-Normal model; **h** Corresponding ground-truth clear image. The figure showcases performance in dense haze conditions, focusing on haze removal completeness and texture preservation

proposed network has only the 2nd smallest number of model parameters under the 'Tiny' structure, and the floating-point computation capacity and runtime are optimal, which can meet the demand of real-time processing.

Notably, the trade-off between accuracy and complexity is evident when comparing the Tiny and Normal variants: on the SOTS dataset, the Normal model achieves a 0.66 dB higher PSNR and 0.0042 higher SSIM than the Tiny model, but at the cost of ~ 3 × higher FLOPs (30.76G vs. 10.28G) and nearly double the inference latency (12.41 ms vs. 6.32 ms). This indicates that the Tiny model prioritizes efficiency, making it suitable for resource-constrained scenarios where real-time performance is critical.

## 4.4 Ablation experiment

To verify the effectiveness of the dehazing module, we analyze the effectiveness of both multi-scale sampling and cross-scale feature aggregation. Among them, the downsampling layer and the dehazed cross-scale feature aggregation model are used as a baseline, and the downsampling layer and the feature aggregation module are added sequentially to compare the performance of each method on NH-HAZE. The results are presented in Table 6, where Down and Fusion represent our downsampling and feature aggregation modules.

**Table 6** The results of ablation experiments with different modules added

| Method | SSIM | PSNR |
|---|---|---|
| Baseline | 0.6322 | 17.34 |
| Baseline + down | 0.6738 | 17.89 |
| Baseline + fusion | 0.6893 | 18.71 |
| OURS-TINY | **0.7034** | **20.34** |

The bold text indicates the 'best-performing results' in the ablation experiments, meaning the final optimal values of the model performance (SSIM, PSNR) after gradually adding key modules and continuously improving performance

The results show that the SSIM is improved by 0.0416 after adding the multi-scale downsampling layer, while the model effect is improved by 0.0571 after adding cross-scale feature aggregation, and the model reaches the optimal effect when both are added simultaneously. It is suggested that the dehazing performance can be significantly improved by cross-size feature aggregation of the dehazed features.

To further investigate the effect of the convolutional kernel size in the proposed M3S module on the dehazing quality and model inference speed, we tried different convolutional kernel sizes in the 'Tiny' model and performed comparative

**Table 7** The ablation experiments with different convolutional kernel sizes in the M3S module

| Kernel size | SSIM | PSNR | Latency (ms) |
|---|---|---|---|
| 1, 3, 5 | 0.6896 | 19.88 | 6.02 |
| 3, 5, 7 | 0.6965 | 19.93 | 6.15 |
| 3, 5, 9 | 0.7003 | 20.13 | 6.23 |
| 5, 7, 9 | 0.7034 | 20.34 | 6.32 |

experiments on the NH-HAZE dataset. The results are shown in Table 7. The experimental results show that the dehazing effect is best when the convolution kernel size is set to 5, 7 and 9, with no significant increase in inference time. Since we use spatially separable convolution, there is spatial locality for vertical or horizontal features of columnar objects, which means that the object features are continuous in a certain direction, so larger convolution kernels are more conducive to extracting the continuous features of the object in the image.

### 4.5 Discussion on limitations

While the proposed method achieves competitive performance on benchmark datasets, it is important to acknowledge its current limitations. First, in extremely dense haze conditions (e.g., visibility < 50 m in Dense-HAZE), the model occasionally exhibits incomplete haze removal, particularly in regions with severe texture occlusion, as the structural information guidance may be weakened by excessive haze particle scattering. Second, the generalization to diverse real-world haze patterns (e.g., fog mixed with dust or humidity variations) is constrained by the limited diversity of existing real hazy datasets (NH-HAZE and Dense-HAZE), which cover relatively specific geographic and climatic conditions. Third, the model relies on paired hazy-clear training data, which are scarce for rare weather events (e.g., sudden fog bursts), limiting its adaptability to such scenarios. These limitations highlight the need for future work on robust structural feature extraction under extreme haze, expansion of multi-scenario real-world datasets, and exploration of unsupervised/semi-supervised learning to reduce dependence on paired data.

## 5 Conclusion

To address the challenges of incomplete detail recovery and lack of some structural information in the process of dehazing outdoor scenes, we propose a single-image dehazing algorithm based on cross-scale feature aggregation with structural information guidance. Aiming to address the problem of difficulty in image detail feature extraction, an image downsampling module based on cross-scale information aggregation is constructed. The module first extracts rich image features by different sizes of dilated convolution and strip convolution and enhances them using the channel attention and pixel attention mechanisms. Then, the downsampling process is performed by SPD-Conv, while the cross-size residual concatenation is introduced to compensate for the detailed information in the deep features. Since outdoor scene images contain rich structural information and compositional patterns, we established the M3S module for image dehazing based on structural information guidance. At the same time, partial convolution is used in the dehazing process to reduce the interference of redundant features and computational complexity. In the image reconstruction stage, to compensate for the lack of deep and shallow features in the global and local information levels of the image, we perform aggregation enhancement of different scales of dehazing features to improve the saliency of dehazing image features with different resolutions. The experimental results show that the proposed dehazing network has SSIM with PSNR scores of 0.9911 and 35.94 dB on the SOTS dataset. The SSIM and PSNR scores on the NH-HAZE dataset are 0.7207 and 21.12 dB. In terms of model operation efficiency, the computational cost of constructing the Tiny model is 10.28G FLOPs, and the inference time is only 6.32 ms, which has the ability of lightweight deployment. The Tiny and Normal variants are designed to cater to different application demands. The Tiny model is preferred in scenarios where real-time performance and low computational resources are paramount, such as edge devices (e.g., surveillance cameras, UAVs) or mobile platforms, where its 6.32 ms latency and 10.28G FLOPs enable continuous operation under hardware constraints—even with a slight accuracy reduction (e.g., 35.28 dB PSNR on SOTS vs. 35.94 dB for Normal). In contrast, the Normal model is better suited for high-precision tasks with sufficient computational resources (e.g., desktop-based image post-processing), where the improved detail restoration justifies the higher complexity. How to improve the adaptability of the network further to meet the practical needs of all-weather dehazing in outdoor scenarios is the focus of our future research work.

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**Author contributions** ZF.Z and RY.W framed the paper structure and wrote core content. XL.Y and ZH.Z conducted in-depth data analysis and designed experimental protocols. Z.L and WB.F collected literature and wrote the review section.QL.Z and ZH.W prepared the figures and tables. K.M and M.W verified the experimental results. L.Y formattedthe final manuscript.

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**Data availability** Data will be made available on reasonable request.

## Declarations

**Conflict of interest** The authors declare no conflict of interests.

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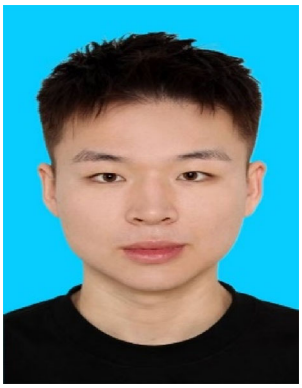

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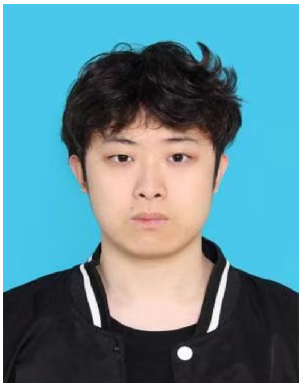

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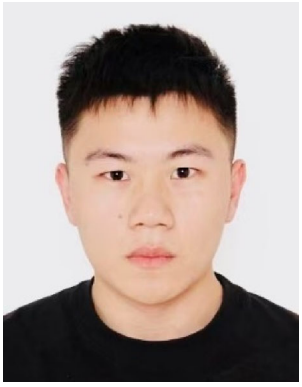

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